# Testing

Two suites, for two different jobs.

| Command | What runs | Needs | Takes |
|---|---|---|---|
| `npm test` | The rules, as plain functions | nothing | ~2 seconds |
| `npm run test:integration` | The API against a real Postgres | Docker | ~30 seconds |
| `npm run test:all` | Both | Docker | ~35 seconds |
| `npm run test:db:stop` | Removes the test database container | Docker | — |

167 tests today: 83 unit, 84 integration.

## The rules ( `src/**/*.test.ts` )

Pure functions with no database: free slots and opening hours, how a visit was paid, session lengths, phone numbers, receipts, who may use which door. Fast enough to keep running while you work ( `npm run test:watch` ).

These exist because the awkward parts of this system are arithmetic — a buffer around an appointment, a mixed payment that must add up, a clock that changes twice a year. Arithmetic is cheap to test and expensive to get wrong.

## The API ( `tests/*.test.ts` )

The route handlers, against a real Postgres in a container, with the session mocked to whichever role the test is about. Eleven files, of which the ones worth knowing about are:

- **`booking-api.test.ts`** — booking, clashes, opening hours, buffers, time off, moving, cancelling, and who may use the shop's diary.
- **`customer-api.test.ts`** — signing up, what a customer may book, the notice period, the limit on waiting bookings, cancelling their own and not others', and the wall between customer and shop.
- **`money-api.test.ts`** — a visit rung up at the till through the drawer, the day journal and the period report: cash, card, InstaPay, mixed, tips, expenses, advances.
- **`auth.test.ts`** — signing in by email or phone, and how long a login lasts.
- **`checkout-api.test.ts`** , **`queue-api.test.ts`** , **`diary-summary-api.test.ts`** — ringing a booking up, the live queue, and the day at a glance.

- `barber-app.test.ts`, `barber-visibility.test.ts` — a barber's own day and earnings, and who the customer app is allowed to show.
- `offers-api.test.ts`, `push-api.test.ts`, `reviews-api.test.ts` — offers, notifications and reminders, and reviews: who may leave one, one per visit, and the owner's power to hide but not delete.
- `loyalty-api.test.ts` — stamps counted from paid visits only, a card that cannot be cashed in twice, and the proof that giving a reward moves no money.
- `insights-api.test.ts` — the owner's read of the shop: the window, who may ask, and the proof that a long-standing customer is never counted as a new face.
- `segments-api.test.ts` — the groups, and the guardrails on writing to one: who may send, once every twelve hours, and only to people who asked for notifications.
- `referrals-api.test.ts` — a code that survives being written down badly, a referral earned only by a friend who actually came in, a reward given once, and a birthday greeting that goes out once a year.

A mock cannot tell you that two people took the same slot, or that the drawer and the report disagree. These can.

The database is a throwaway container ( `escape-test-db`, port 55499), started on the first run and reused afterwards. Every table is emptied between tests, and the connection string is built in `tests/db.ts` — the shop's real database is never reachable from a test.

## Writing another one

- A new rule starts as a function in `src/lib` with a unit test, before any screen.
- A new endpoint gets an integration test covering what it does **and** who is refused. Every route was written with both, and one of those refusal tests is what caught a customer being able to open the shop's own appointments screen.
- Reach for real rows over mocks. The bugs live in what the database actually did.

## What is not covered yet

- The screens themselves: no browser tests. Checked by hand for now.
- The Excel and PDF exports beyond the receipt text.
- The warehouse and payroll screens, which predate this suite and are covered only through the money tests they feed.